
Explainable Decision-Making in Non-Stationary Environments via LLM-Guided Temporal Logic Reasoning

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Abstract

Decision-making systems deployed in non-stationary environments increasingly rely on adaptive planning algorithms that continuously revise internal models, policies, or beliefs. Although such adaptivity improves robustness, it poses a fundamental challenge to explainability. Existing explanatory methods implicitly assume a fixed planning representation and therefore cannot account for the shifting decision rationales that arise across adaptations. We introduce a formal explainability framework for adaptive planning in non-stationary environments, centered on the abstraction of a planning snapshot, which is a time-indexed and self-contained representation of a planner’s internal state that fully determines its behavior at each decision epoch. We identify sufficient compatibility conditions under which an adaptive planner admits snapshot-based explanation and define a structured taxonomy of explanation queries grounded in probabilistic computation tree logic (PCTL) specifications evaluated directly on snapshots. As explanations are grounded in the planner’s own internal representation rather than post-hoc approximations, they are faithful to the planner’s actual reasoning and remain consistent with its adaptation over time. We **[TODO]** show how the framework instantiates across different adaptation paradigms and provide a detailed empirical evaluation using ADA-MCTS on a real-world paratransit vehicle dispatching problem, demonstrating that the framework produces accurate and time-aware explanations.

1 Introduction

Real-world decision-making systems often operate in environments where data distributions and system dynamics evolve over time. Ditzler et al. [2015] In many practical scenarios, intrinsic non-stationarity arises from factors such as seasonality, shifts in user behavior, or fluctuations in traffic conditions. Under such circumstances, non-adaptive models trained under the assumption of a stationary environment are inevitably suboptimal or, even worse, fail entirely when facing conditions that deviate from their prior assumptions. Thus, adaptive planning algorithms are specifically developed to account for these situations.

While adaptive planning improves robustness in non-stationary environments, the decision-making behavior of adaptive planners becomes increasingly difficult to interpret as the environment evolves. Despite recent advances in explainable artificial intelligence (XAI), most existing explanation methods are implicitly designed under the assumption that the underlying decision-making algorithm is stationary Baier and Kaisers [2021]. Explanations are typically generated with respect to a fixed planning representation such as a static policy, value function, or search tree **[TODO]** **[TODO]**: more XAI citations, and are therefore expected to remain consistent over time. However, this assumption

no longer holds for adaptive planners in non-stationary environments. When a planner’s internal state evolves across adaptation stages, identical actions observed at different time points may arise from fundamentally different planning objectives, and explanations that ignore this temporal dimension become unreliable or misleading.

Despite these challenges, adaptive planning algorithms share a common structure that can be leveraged for explainability. Existing non-stationary planners adapt in different ways: some revise explicit environment models such as time-indexed MDP estimates Luo et al. [2024], some continually update learned dynamics or latent context representations Fu and Gao [2025], and others primarily adjust planner priors or exploration biases Pettet et al. [2022]. While these approaches differ, at each decision epoch, the planner’s behavior is fully determined by a time-indexed internal state that governs how future outcomes are simulated and evaluated. We refer to this as a planning snapshot. This abstraction provides a unified foundation for explainability across diverse adaptation mechanisms. Rather than requiring a separate explanation method for each type of adaptive algorithm, we can ground explanations in the planning snapshot at the relevant time point, capturing both what the planner decided and what internal state led to that decision.

Building on this abstraction, we introduce a formal explainability framework for adaptive planning in non-stationary environments. We first identify sufficient compatibility conditions under which an adaptive planner admits snapshot-based explanation, establishing the scope of algorithms to which the framework applies. We then define categories of explanation queries grounded in probabilistic computation tree logic (PCTL) specifications evaluated directly on planning snapshots. Because explanations are derived from the planner’s own internal representation at each decision epoch rather than from post-hoc approximations of its behavior, they are faithful to the planner’s actual reasoning and remain temporally consistent as the planner adapts over time. **[TODO]**The framework translates natural language user queries into formal PCTL specifications through two strategies, one template-based and one that leverages large language models for flexible logic synthesis, both sharing a common downstream evaluation and evidence-based explanation generation process.

The main contributions of this work are as follows. First, we introduce the planning snapshot abstraction and identify formal compatibility conditions that characterize what kind of adaptive planning algorithms could be used by the snapshot-based explanation, providing a principled foundation for explainability in non-stationary environments. Second, we develop an explainability framework that grounds user queries in PCTL specifications evaluated on planning snapshots, ensuring faithfulness and temporal consistency of the generated explanations. Third, we provide a detailed empirical evaluation by instantiating the framework on ADA-MCTS applied to a real-world paratransit vehicle dispatching problem, demonstrating that the framework produces accurate and time-aware explanations that capture behavioral shifts induced by environmental change.

2 Related Work

2.1 XAI for Decision-Making under Stationary Assumptions

Baier and Kaisers [2021]

[TODO]TODO: add 1-2 more

2.2 Adaptive and Non-Stationary Planning Algorithms

Non-stationary planning algorithms differ primarily in how they represent and update their understanding of the environment. Based on this distinction, existing approaches can be broadly grouped into three categories.

2.2.1 Model-Based Tree Search Planners

Model-based tree search planners combine online tree search with an explicit forward model defined over the original state space of the problem domain. The forward model may be learned from data or provided as a simulator. ADA-MCTS Luo et al. [2024] learns a Bayesian Neural Network (BNN) to estimate transition dynamics $\hat{P}_t(s'|s, a)$, periodically retraining the model on recent observations and using the resulting uncertainty estimates to adapt its search strategy. PA-MCTS Pettet et al. [2022] instead relies on a ground-truth environment simulator for forward search, augmenting the standard

UCT selection rule with Q-value estimates from a pre-trained offline policy to accelerate convergence after environmental changes. LiZero Zhang and Lan [2025] also operates with a true simulator, but transfers value estimates across consecutive non-stationary tasks via Lipschitz-bounded adaptive UCT to enable faster re-planning. Despite differences in how the model or prior knowledge is incorporated, these methods share a common structure in which a forward model over the original state space is queried within an online tree search procedure to produce the decision policy.

2.2.2 Tabular Model-Based Reinforcement Learning

A second category of non-stationary planning algorithms maintains explicit tabular estimates of transition probabilities and reward over a finite state space, deriving policies through optimistic or no-regret planning rather than tree search. SWUCRL2-CW Cheung et al. [2020] computes empirical transition estimates $\hat{P}_t(s'|s, a)$ and reward estimates $\hat{r}_t(s, a)$ from a sliding window of recent observations, applying confidence widening to account for potential non-stationarity within the window. The MASTER framework Wei and Luo [2021], when instantiated with a model-based base algorithm such as UCRL, maintains multiple algorithm instances at geometrically increasing time scales and selects among them via a bandit mechanism. Each UCRL instance builds its own empirical model from the allocated data. Even-Dar et al. [2009] address a setting where the transition dynamics P are known and fixed but rewards change adversarially, adapting the policy via no-regret algorithms applied to the known MDP. Despite differences in how they handle non-stationarity, these methods share a common structure: an explicit transition model defined over a finite state space, from which the policy is derived.

2.2.3 Model-Free Adaptive Reinforcement Learning

A third category of algorithms adapts to non-stationarity without ever constructing a transition model. These methods maintain only value estimates such as Q-values or parameterized value functions, and adapt them through mechanisms such as periodic restarts or context switching. RestartQ-UCB Mao et al. [2022] periodically discards and re-learns Q-value estimates at fixed intervals. LSVI-UCB-Restart Zhou et al. [2022] applies the same restart principle to least-squares value iteration with linear function approximation, directly estimating the Q-function without recovering the underlying transition parameters. Context Q-Learning Padakandla et al. [2020] uses change-point detection on state-reward trajectories to identify environment shifts and maintains separate Q-value estimates for each detected context. These algorithms achieve strong regret guarantees for non-stationary MDPs, but by design they learn value estimates without learning transition dynamics.

The architectural distinction between these categories, namely whether an algorithm maintains an explicit and queryable forward model, has direct implications for explainability, as we formalize in Section 3.2 Planning Snapshot.

2.3 Formal Methods for Explainability

[TODO]TODO: existing XAI with formal methods – why PCTL is used

3 Problem Formulation

3.1 Non-Stationary Adaptive Planning

A non-stationary environment is one in which the underlying dynamics governing state transitions and rewards change over time. We formalize this as a non-stationary Markov decision process (MDP), defined by a tuple (S, A, T_t, R_t, γ) , where S is the state space, A is the action space, $\gamma \in [0, 1)$ is the discount factor, and T_t and R_t are the transition function and reward function at time t . Unlike stationary MDPs where T and R are fixed, here these functions may vary across decision epochs due to factors such as shifting demand patterns or changing environmental conditions.

An adaptive planner operates over a sequence of decision epochs $t = 1, 2, \dots$ and periodically revises its internal representation of the environment in response to newly observed data. As discussed in Section 2.2, the specific revision mechanism varies across algorithms, but all adaptive planners share the property that their behavior at any given epoch is determined by a time-indexed internal state that reflects accumulated observations and model revisions up to that point.

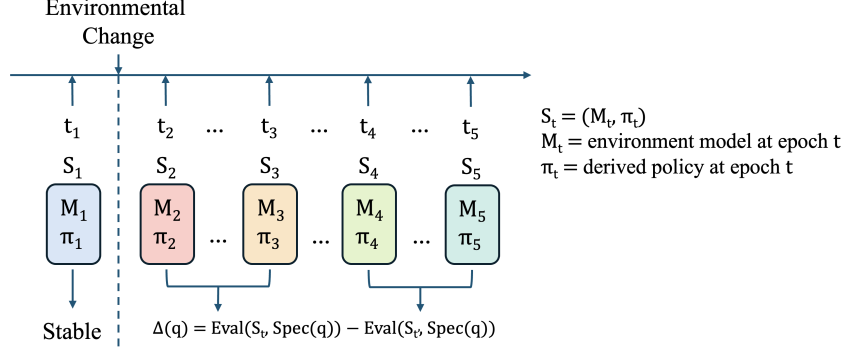


Figure 1: Planning snapshots over time in a non-stationary environment. Each snapshot $S_t = (M_t, \pi_t)$ captures the planner’s environment model and derived policy at decision epoch t . After an environmental change, the planner continuously adapts, producing a sequence of evolving snapshots. Cross-snapshot evaluation compares formal specification results between snapshots to quantify behavioral shifts.

135 3.2 Planning Snapshot

136 Although adaptive planners differ in their revision mechanisms, they share a common structural
 137 property: at each decision epoch, the planner’s behavior is fully determined by a time-indexed internal
 138 state. We formalize this observation through the abstraction of a planning snapshot, illustrated in
 139 Figure 2.

140 subsubsectionDefinition 1 (Planning Snapshot)

141 A planning snapshot at decision epoch t is a tuple $S_t = (M_t, \pi_t)$, where M_t is the planner’s
 142 environment model encoding its current beliefs about the transition dynamics and reward structure,
 143 and π_t is the derived policy computed from M_t , either explicitly (e.g., through policy optimization)
 144 or implicitly (e.g., through online tree search). The snapshot is self-contained in the sense that it fully
 145 determines the planner’s decision-making behavior at epoch t without requiring access to any prior
 146 observations or historical states.

147 3.2.1 Snapshot Difference

148 Two snapshots S_t and $S_{t'}$ differ when the planner’s internal state has changed between epochs t' and
 149 t , typically as a result of a model update triggered by newly observed data. A difference in snapshots
 150 implies that the planner’s behavior may have changed, meaning the same query about the planner’s
 151 reasoning may yield a different answer at different epochs. This temporal variation is precisely what
 152 the explanation framework must account for.

153 3.2.2 Definition 2 (Snapshot Compatibility)

154 An adaptive planner is snapshot-compatible if it satisfies the following conditions:

155 (C1) *Time-indexed internal state*. At each decision epoch t , the planner maintains an internal state
 156 that can be represented as a planning snapshot $S_t = (M_t, \pi_t)$.

157 (C2) *Simulatable forward model*. The environment model M_t induces a well-defined MDP that
 158 supports forward simulation from any reachable state.

159 (C3) *Extractable atomic propositions*. Boolean predicates on states of the induced MDP can be
 160 defined and evaluated, enabling formal property specification.

161 A planner satisfying these conditions admits snapshot-based explanation, as the induced MDP at
 162 each epoch can serve as the formal structure on which temporal logic properties are evaluated. The
 163 three categories of adaptive algorithms discussed in [TODO]Section 2.2 all satisfy these conditions:
 164 [TODO]finish

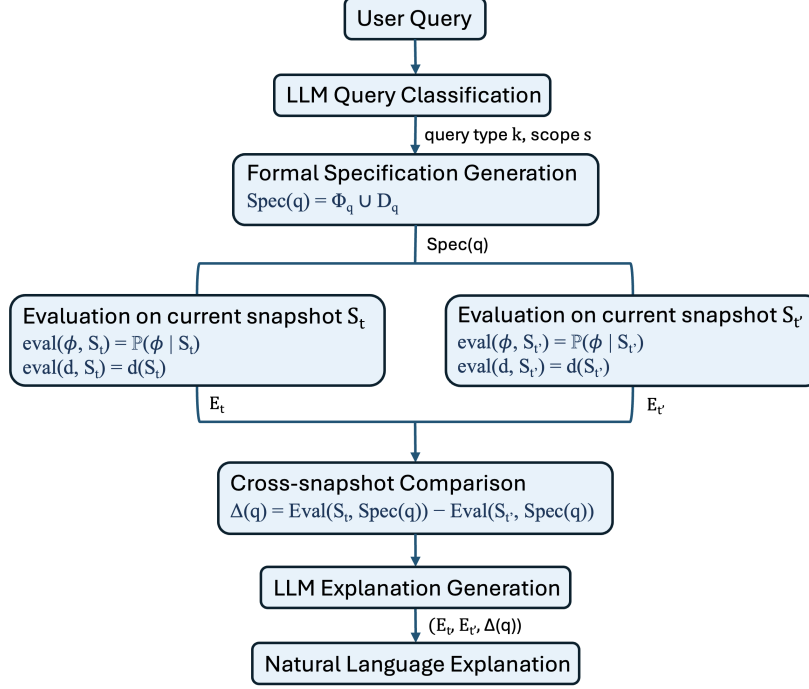


Figure 2: Formal pipeline of the explanation framework. A user query is classified by an LLM into query type k and scope s , which are used to generate the formal specification $Spec(q) = \Phi_q \cup D_q$. The specification is evaluated independently on the current snapshot S_t and the previous snapshot $S_{t'}$, producing evidence sets containing PCTL results and derived metric values. Cross-snapshot comparison computes the temporal difference $\Delta(q)$, which an LLM synthesizes into a natural language explanation.

3.3 Explanation Problem

In stationary settings, an explanation for a planner’s decision can be generated with respect to a single, fixed planning representation. In non-stationary settings, this is insufficient. Because the planner’s internal state evolves over time, an explanation must account for not only what the planner decided but also how its reasoning has changed relative to a previous state. We formalize this as follows.

3.3.1 Definition 3 (Explanation Task)

Given a snapshot-compatible adaptive planner, a pair of planning snapshots $(S_t, S_{t'})$ where $t' < t$, and a natural language user query q , the explanation task is to produce a natural language explanation $E(q, S_t, S_{t'})$ that answers q with respect to the behavioral difference between S_t and $S_{t'}$.

This formulation reflects the fact that all explanation queries in our framework are grounded in cross-snapshot comparison. The user does not query a single snapshot in isolation, but instead asks about the planner’s behavior in the context of how it has adapted over time.

4 Method

4.1 Framework Overview

To explain the decision-making behavior of adaptive planners in non-stationary environments, we propose an explanation framework that bridges natural language user queries with formal temporal logic analysis grounded in the planner’s internal state. The detailed pipeline is illustrated in Figure 3.

Given a natural language query, the framework proceeds in four stages. First, an LLM classifies the query into a predefined query type k and a scope parameter s that determines whether the analysis

184 targets a specific request or an entire decision session (Section 4.2). Second, the classified query type
 185 is used to generate a formal specification $Spec(q)$ consisting of PCTL formulas and derived metrics,
 186 through a hierarchical construction that selects relevant atomic propositions and temporal operators
 187 based on the query semantics (Section 4.3). Third, the formal specification is evaluated independently
 188 on the current planning snapshot S_t and the previous snapshot $S_{t'}$, and the results are compared to
 189 quantify how the planner’s behavior has shifted across model updates (Section 4.4). Finally, the
 190 cross-snapshot evaluation results are synthesized by an LLM into a natural language explanation that
 191 explicitly contrasts the planner’s reasoning before and after environmental change (Section 4.5).

192 4.2 Query Classification

193 [TODO]matching k , s to c , h in the figure

194 The first stage of the framework maps a natural language user query q into a structured representation
 195 that determines the subsequent formal analysis. Specifically, an LLM classifies each query into two
 196 components: a query type $k \in \{1, \dots, K\}$ and a scope parameter $s \in \{micro, macro\}$. The query
 197 type k captures the explanatory intent of the query. Each type corresponds to a distinct category
 198 of questions that a user may ask about the planner’s behavior, such as why a particular action was
 199 selected and how a specific metric has changed across model updates. The set of query types is
 200 defined at the domain level when the framework is instantiated on a specific adaptive planner, and
 201 each type is associated with a fixed formal specification. The scope parameter s determines the
 202 granularity of the analysis. A micro-scope query targets a specific decision instance (e.g., a single
 203 request assignment), whereas a macro-scope query targets aggregate behavior over the entire decision
 204 session (e.g., overall service reliability). This distinction affects how the formal specification is
 205 evaluated on the planning snapshot, as micro-scope queries condition the evaluation on a particular
 206 state, while macro-scope queries evaluate from the root state of the MDP, covering the full decision
 207 session.

208 In practice, the classification is performed by prompting an LLM with the user query with descriptions
 209 of the available query types. Because the set of query types is predefined and finite, the classification
 210 task is a standard multi-class assignment, and the LLM serves as a flexible natural language interface
 211 that accommodates varied phrasings of the same underlying explanatory intent.

212 4.3 Formal Specification Generation

213 Given a classified query type k from Section 4.2, the framework defines a formal specification
 214 that precisely defines the set of temporal logic properties and quantitative metrics to be evaluated
 215 on the planning snapshot. The construction follows a hierarchical procedure illustrated in Figure
 216 4, proceeding from domain-level definitions through query-specific selection to the final formal
 217 specification.

218 4.3.1 Atomic Propositions

219 Let AP denote the full set of atomic propositions extractable from a planning snapshot S . Each
 220 atomic proposition $ap \in AP$ is a Boolean predicate evaluated on states of the snapshot-induced
 221 MDP. We categorize AP into semantically disjoint categories $AP = AP_1 \cup AP_2 \cup \dots \cup AP_m$,
 222 where each category groups propositions that characterize a common aspect of the planner’s behavior.
 223 For instance, in the paratransit vehicle dispatching problem, one may define AP_{out} for service
 224 outcome propositions, AP_{delay} for delay-related propositions, and AP_{ops} for operational constraint
 225 propositions. The specific partition is determined at instantiation time and depends on the structure
 226 of the planning domain.

227 4.3.2 Temporal Operators

228 Let $T = \{F, X, G, U\}$ denote the standard PCTL temporal operators, where F (finally) asserts
 229 that a property holds at some future state, X (next) asserts it holds at the immediate successor, G
 230 (globally) asserts it holds along the entire path, and U (until) asserts that one property holds until
 231 another becomes true. We define four temporal horizon categories H as single-step, finite-horizon,
 232 infinite-horizon, and conditional-horizon, with a bijection of $H \rightarrow T$.

4.3.3 Derived Metrics

In addition to PCTL formulas, certain explanation queries require quantitative metrics that are computed directly from the snapshot state rather than through probabilistic model checking. Let $D = D_1 \cup D_2 \cup \dots \cup D_m$ denote the full set of derived metrics, organized into categories that are aligned with the atomic proposition categories. Each derived metric $d \in D$ is a real-valued function evaluated on a planning snapshot, mapping $d(S_t) \rightarrow \mathbb{R}$.

4.3.4 Query-Specific Selection

Given a query of type k with content focus c_q and temporal horizon h_q , the query-specific components are:

$$AP_q = \bigcup_{i \in c_q} AP_i, \quad T_q = \{ \text{op}(h) : h \in h_q \}, \quad D_q = \bigcup_{i \in c_q} D_i.$$

For example, a service reliability query with $c_q = \{\text{out}, \text{delay}\}$ and $h_q = \{\text{single-step, finite-horizon, infinite-horizon}\}$ yields AP_q drawn from service outcome and delay propositions, and $T_q = \{X, F, G\}$.

4.3.5 PCTL Formula Construction

The set of PCTL formulas for query q is constructed via the Cartesian product of the selected atomic propositions and temporal operators. For operators F, X , and G , each pair $(ap, op) \in AP_q \times T_q$ yields a PCTL state formula:

$$\phi_{ap, op} = P_{=?}[op \ ap]$$

which queries the probability that the temporal property $op \ ap$ is satisfied from a given state under the snapshot-induced MDP. For the until operator U , formulas take the form $P_{=?}[ap_1 \cup ap_2]$ where both operands are drawn from AP_q ; the specific pairings are defined per query type.

The complete PCTL formula set is:

$$\Phi_q = \{ \phi_{ap, op} : ap \in AP_q, op \in T_q \}.$$

4.3.6 Formal Specification

The formal specification for query q is the union of the PCTL formula set and the derived metrics set:

$$Spec(q) = \Phi_q \cup D_q$$

This specification fully determines the set of formal properties to be evaluated on any planning snapshot. Because the construction is deterministic given the query type k , the same query always produces the same specification regardless of when it is issued, ensuring reproducibility of the formal grounding stage.

4.4 Cross-snapshot Evaluation

Given a formal specification $Spec(q)$ generated in Section 4.3, the framework evaluates it on both the current planning snapshot S_t and the previous snapshot $S_{t'}$ to produce comparable evidence sets that capture the planner's behavior before and after a model update.

4.4.1 Snapshot Evaluation

For a single planning snapshot S_t , evaluation proceeds over the two components of the formal specification. For each PCTL formula $\varphi \in \Phi_q$, a probabilistic model checker evaluates the formula on the MDP induced by S_t :

$$eval(\phi, S_t) = \mathbb{P}(\phi | S_t) \in [0, 1]$$

For each derived metric $d \in D_q$, the corresponding value is computed directly from the snapshot state:

$$eval(d, S_t) = d(S_t) \in \mathbb{R}$$

269 The complete evaluation result for snapshot S_t is the evidence set:

$$E_t = \{eval(f, S_t) : f \in Spec(q)\}$$

270 which contains one numerical value for each element of the formal specification. The same procedure
271 is applied to the previous snapshot $S_{t'}$, yielding evidence set $E_{t'}$.

272 4.4.2 Cross-Snapshot Comparison

273 The behavioral shift between two snapshots is quantified by the element-wise difference between
274 their evidence sets:

$$\Delta(q) = E_t - E_{t'} = Eval(S_t, Spec(q)) - Eval(S_{t'}, Spec(q))$$

275 Each component of $\Delta(q)$ indicates how a specific formal property has changed across the model
276 update. A positive value indicates an increase under the current model, while a negative value
277 indicates a decrease. Together, the triple $(E_t, E_{t'}, \Delta(q))$ forms the complete evidence package
278 passed to the explanation generation stage.

279 4.4.3 Scope-Dependent Evaluation

280 The scope parameter s from Section 4.2 determines the scope of the evaluation. Under micro scope,
281 the MDP is initialized at the specific state corresponding to the target decision instance (e.g., the state
282 associated with a particular request assignment), and all PCTL formulas are evaluated from that state.
283 Under macro scope, the evaluation is performed from the root state of the MDP, thereby covering the
284 full decision session.

285 4.5 Explanation Generation

286 The final stage synthesizes the evaluation results into a natural language explanation. An LLM
287 receives a structured prompt containing the environment context, the user’s original query, the
288 evaluation results E_t and $E_{t'}$ for all candidate actions under both snapshots. The LLM generates an
289 explanation that addresses the query with explicit reference to the planner’s behavioral shift.

290 The generated explanations follow a contrastive structure that mirrors the cross-snapshot evidence.
291 For each candidate action relevant to the query, the explanation compares its property values under
292 the previous model and the current model, then concludes by answering the user’s query based on the
293 observed differences. The LLM is constrained to reference only the values provided in the evidence,
294 ensuring that every claim in the explanation is traceable to a formal evaluation result.

295 5 Experiments and User Study

296 5.1 Setup

297 paratransit domain, snapshots, queries, baselines B1-B4

298 5.2 Faithfulness

299 Numerical fidelity main table

300 5.3 Temporal Faithfulness

301 Δ -Recovery results table

302 5.4 Robustness (5.3 sub)

303 Numerical consistency + Claim Jaccard

304 5.5 Ablation: Reference Snapshot Selection

305 A4(adaptive vs fixed reference) only

306 5.6 Plausibility (User Study)

307 Likert results, explicitly separated from faithfulness

308 5.7 Efficiency (appendix)

309 Timing breakdown table

310 6 Discussions and Limitations

311 7 Conclusion

312 8 Citations, figures, tables, references

313 These instructions apply to everyone.

314 8.1 Citations within the text

315 The `natbib` package will be loaded for you by default. Citations may be author/year or numeric, as
316 long as you maintain internal consistency. As to the format of the references themselves, any style is
317 acceptable as long as it is used consistently.

318 The documentation for `natbib` may be found at

319 `http://mirrors.ctan.org/macros/latex/contrib/natbib/natnotes.pdf`

320 Of note is the command `\citet`, which produces citations appropriate for use in inline text. For
321 example,

322 `\citet{hasselmo}` investigated\dotso

323 produces

324 Hasselmo, et al. (1995) investigated...

325 If you wish to load the `natbib` package with options, you may add the following before loading the
326 `neurips_2026` package:

327 `\PassOptionsToPackage{options}{natbib}`

328 If `natbib` clashes with another package you load, you can add the optional argument `nonatbib`
329 when loading the style file:

330 `\usepackage[nonatbib]{neurips_2026}`

331 As submission is double blind, refer to your own published work in the third person. That is, use “In
332 the previous work of Jones et al. [4],” not “In our previous work [4].” If you cite your other papers
333 that are not widely available (e.g., a journal paper under review), use anonymous author names in the
334 citation, e.g., an author of the form “A. Anonymous” and include a copy of the anonymized paper in
335 the supplementary material.

336 8.2 Footnotes

337 Footnotes should be used sparingly. If you do require a footnote, indicate footnotes with a number¹
338 in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote
339 with a horizontal rule of 2 inches (12 picas).

340 Note that footnotes are properly typeset *after* punctuation marks.²

¹Sample of the first footnote.

²As in this example.

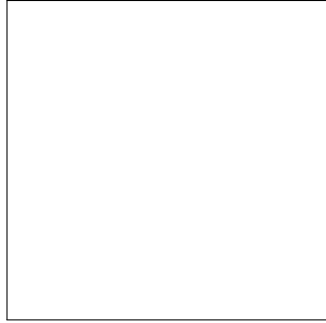


Figure 3: Sample figure caption. Explain what the figure shows and add a key take-away message to the caption.

Table 1: Sample table caption. Explain what the table shows and add a key take-away message to the caption.

Part		
Name	Description	Size (μm)
Dendrite	Input terminal	≈ 100
Axon	Output terminal	≈ 10
Soma	Cell body	up to 10^6

8.3 Figures

All artwork must be neat, clean, and legible. Lines should be dark enough for reproduction purposes. The figure number and caption always appear after the figure. Place one line space before the figure caption and one line space after the figure. The figure caption should be lower case (except for the first word and proper nouns); figures are numbered consecutively.

You may use color figures. However, it is best for the figure captions and the paper body to be legible if the paper is printed in either black/white or in color.

8.4 Tables

All tables must be centered, neat, clean, and legible. The table number and title always appear before the table. See Table 1.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for the first word and proper nouns); tables are numbered consecutively.

Note that publication-quality tables *do not contain vertical rules*. We strongly suggest the use of the `booktabs` package, which allows for typesetting high-quality, professional tables:

<https://www.ctan.org/pkg/booktabs>

This package was used to typeset Table 1.

8.5 Math

Note that display math in bare TeX commands will not create correct line numbers for submission. Please use LaTeX (or AMSTeX) commands for unnumbered display math. (You really shouldn't be using `$$` anyway; see <https://tex.stackexchange.com/questions/503/why-is-preferable-to> and <https://tex.stackexchange.com/questions/40492/what-are-the-differences-between-align-equation-and-displaymath> for more information.)

365 8.6 Final instructions

366 Do not change any aspects of the formatting parameters in the style files. In particular, do not modify
367 the width or length of the rectangle the text should fit into, and do not change font sizes. Please note
368 that pages should be numbered.

369 9 Preparing PDF files

370 Please prepare submission files with paper size “US Letter,” and not, for example, “A4.”

371 Fonts were the main cause of problems in the past years. Your PDF file must only contain Type 1 or
372 Embedded TrueType fonts. Here are a few instructions to achieve this.

- 373 • You should directly generate PDF files using `pdflatex`.
- 374 • You can check which fonts a PDF files uses. In Acrobat Reader, select the menu
375 Files>Document Properties>Fonts and select Show All Fonts. You can also use the program
376 `pdf fonts` which comes with `xpdf` and is available out-of-the-box on most Linux machines.
- 377 • `xfig` “patterned” shapes are implemented with bitmap fonts. Use “solid” shapes instead.
- 378 • The `\bbold` package almost always uses bitmap fonts. You should use the equivalent AMS
379 Fonts:

380 `\usepackage{amsfonts}`

381 followed by, e.g., `\mathbb{R}`, `\mathbb{N}`, or `\mathbb{C}` for $\mathbb{R}$, $\mathbb{N}$ or $\mathbb{C}$. You can also
382 use the following workaround for reals, natural and complex:

```
383 \newcommand{\RR}{\mathbb{R}} %real numbers
384 \newcommand{\Nat}{\mathbb{N}} %natural numbers
385 \newcommand{\CC}{\mathbb{C}} %complex numbers
```

386 Note that `amsfonts` is automatically loaded by the `amssymb` package.

387 If your file contains type 3 fonts or non embedded TrueType fonts, we will ask you to fix it.

388 9.1 Margins in L^AT_EX

389 Most of the margin problems come from figures positioned by hand using `\special` or other
390 commands. We suggest using the command `\includegraphics` from the `graphicx` package.
391 Always specify the figure width as a multiple of the line width as in the example below:

```
392 \usepackage[pdftex]{graphicx} ...
393 \includegraphics[width=0.8\linewidth]{myfile.pdf}
```

394 See Section 4.4 in the graphics bundle documentation ([http://mirrors.ctan.org/macros/](http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf)
395 [latex/required/graphics/grfguide.pdf](http://mirrors.ctan.org/macros/latex/required/graphics/grfguide.pdf))

396 A number of width problems arise when L^AT_EX cannot properly hyphenate a line. Please give LaTeX
397 hyphenation hints using the `\-` command when necessary.

398 References

399 Hendrik Baier and Michael Kaisers. Towards explainable mcts. In *2021 AAAI Workshop on*
400 *Explainable Agency in AI*, volume 178, 2021.

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